EFU-HSW LICENCE v2.0 – RESEARCH AUDIT EDITION

Identifier: EFU-A/HSW-2026-01-R02-EN
Status: Research Material – Open for Critical Review (Research Phase)
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Date: January 2026
Licence: EFU Universal Licence v1.0 (CC BY 4.0 compatible)

ABSTRACT

This document constitutes the **theoretical extension of the EFU (Human Flux Unit)** concept into the *Human Software* domain — education, training, and knowledge transmission. Its aim is to model the physical, energetic, and cognitive expenditures of educational systems on a unified **EFU-HSW (Human Software)** scale.

The licence remains strictly a **research-phase construct**, not a normative or educational regulation. It is a conceptual experiment for analysing the *metabolism of knowledge transfer* and the conditions of *educational sovereignty* in flux terms.

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0. Research Status and Scope

[HYPOTHESIS] EFU-HSW is not a curricular, accreditation, or certification instrument, but an experimental **flux-based analytical framework**.

[PROTOCOL] It is intended exclusively for research and falsification, providing a replicable method to express educational and cognitive processes in physical terms of throughput and entropy.

I. Core Concepts and Components

[HYPOTHESIS] Each educational system exhibits identifiable flux components that can be expressed in EFU-equivalents per student-year.

Defined modules:

- **EFU-E_HSW** – Physical energy consumption (lighting, heating, infrastructure)
- **EFU-M_HSW** – Material flux (books, equipment, amortisation)
- **EFU-C_HSW** – Cognitive energy (learning time × intensity)
- **EFU-K_HSW** – Qualification entropy (re-training, dropout)
- **EFU-T_HSW** – Mobility flux (commuting for learning)

Composite flux balance:

TEFU_HSW = 1.5 C + 1.0 E + 1.2 M + 0.8 T + 2.0 K

II. Educational Sovereignty and Metabolic Dignity

An educational system attains **sovereignty** when the knowledge-generation flux equals or exceeds its total input (TEFU).

Metabolic Dignity (framework-specific term) denotes the minimal educational flux required to sustain the cognitive self-regeneration capacity of society.

SS_HSW = TEFU_total − RTHK_HSW
MROI_HSW = THI_HSW / TEFU_HSW

Where:
RTHK = Regional Knowledge Carrying Capacity
THI = Total Human Intelligence gain

III. Regional Calibration (Research Benchmark)

Country	Student-years (M/yr)	Population (M)	RTHK_HSW	Observation
Finland	0.6	5.5	0.11	Negative TEFU – Sovereign
Estonia	0.15	1.3	0.12	Stable MROI
Japan	10	125	0.08	High-density flux system
USA	50	340	0.15	High energy intensity
Hungary	1.5	9.7	0.15	Flux deficit detected

IV. Epistemological Appendix (EA)

Falsifiability

The EFU-HSW model is refuted if no statistical correlation ($R^2 < 0.2$) is found between total educational flux (TEFU) and learning-performance metrics (e.g. PISA score, THI_HSW).

It is also refuted if systems with lower (negative) TEFU values do not exhibit higher competence or innovation outcomes.

Uncertainty Factors

- Cultural bias in cognitive-intensity assessment
- Limited availability of institutional energy data
- Subjective evaluation of dropout entropy (K-component)
- Temporal fluctuation of SS_HSW (Sovereignty Gap)

V. Critical Argument-Map Audit

This section identifies the core reasoning chains of the EFU-HSW framework, their potential vulnerabilities, and empirically testable falsifiers.

(table unchanged in content, formatting normalized for Word/PDF)

#	Research Statement (Hypothesis)	Logical/Empirical Basis	Possible Vulnerability	Proposed Falsifier (test / measurement)
1	Learning can be described as a physical-energetic flux.	Thermodynamic axiom: no information transfer without energy input.	Information as non-physical phenomenon.	Low-energy learning experiment vs. outcome retention.
2	The 20 kg/day throughput (1 EFU) can be scaled to cognitive flux.	Analogy between biophysical and mental metabolism.	Non-linear scaling of mental effort.	Neuro-metabolic fMRI vs. classroom energy use.
3	EFU-C_HSW ∝ learning time × intensity.	Anderson Learning Time Hypothesis (1985).	Quality not proportional to time.	Regression of scores vs. hours of study.
4	EFU-E_HSW captures physical environment impact on learning.	Building energy records.	External temperature distortion.	Weather-corrected energy-performance regression.
5	EFU-K_HSW (dropout entropy) quantifies information loss.	Shannon-entropy analogy.	Dropout personal reasons ≠ information loss.	Long-term dropout trend analysis.
6	TEFU_HSW represents the system's overall metabolic load.	Linear sum of modules.	Ignoring non-linear interactions.	Monte-Carlo simulation of component interplay.
7	Negative TEFU signifies net knowledge accumulation.	Flux-balance logic.	Knowledge non-conservative quantity.	Correlation with competence indices.
8	SS_HSW < 0 indicates Sovereign state.	Cognitive regeneration > resource use.	High parameter sensitivity.	Regional time-series validations.
9	MROI_HSW ≥ 0.1 defines metabolic efficiency.	THI/TEFU ratio.	THI composition vague.	Empirical GDP-contribution test.
10	Finnish negative TEFU yields higher learning outcomes.	OECD evidence.	Cultural interference.	Longitudinal Finland–Hungary comparison.
11	Cognitive and physical fluxes can substitute each other.	Flux transfer analogy.	Heterogeneous dimensions.	Lighting–performance simulation.
12	EFU scaling permits cross-country comparison.	Normalised by population and student-years.	Structural system differences.	GDP-per capita normalised analysis.
13	EFU-HSW enables long-term MROI projection.	Model structure.	Lack of 20-year data.	Pilot cohort tracking.
14	A Metabolic Dignity threshold exists.	Socio-ecological premise.	Cultural dependency.	Development-vs-EFU ratio analysis.
15	EFU-based licence can integrate into future standards.	Open modular design.	Risk of standardisation trap.	Pilot ISO/EN compatibility test.
16	Temporal fluctuation of SS_HSW is essential for falsifiability.	Systems-theory noise.	Lack of long series.	Five-year panel dataset creation.
17	EFU-T_HSW (transport) is a major flux component.	Infrastructure energy data.	Remote learning bias.	Online/offline ratio tracking.
18	Educational EFU footprint can predict economic growth.	Flux-GDP correlation.	Direction of causality unclear.	Cross-temporal correlation analysis.
19	EFU-HSW does not replace pedagogical assessment but adds objective background.	Method definition.	Over-quantification risk.	Qualitative vs quantitative comparative study.
20	Open-source EFU-HSW licence strengthens Open Education Science.	CC BY 4.0 compatibility.	Jurisdictional interpretation issues.	Licence case analysis.

Research Integrity Note

The **Critical Argument-Map Audit** does not validate, certify, or endorse the EFU-HSW model. Its sole purpose is to expose where and how the framework may be empirically tested, challenged, or falsified.

Each hypothesis listed constitutes an open line of inquiry for subsequent calibration and field research.

This document represents an ongoing research construct. All models, thresholds, coefficients, and classifications are provisional and subject to revision, falsification, or withdrawal.

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